CHAPTER 13

A Production-Like Playground

We're *so* close. In the previous chapter, we configured our application so it's ready to run in production environments, and we pushed our image to Docker Hub. However, before we start setting up cloud-based infrastructure and deploying it there, we're first going to get our sea legs, so to speak, by having a practice run locally.

In this chapter, we'll make use of virtualization technologies—specifically, VirtualBox—to create virtual infrastructure on our local machine that can simulate a production environment. We'll create a single node cluster, capable of running Docker containers. Not only will we deploy our application, we'll also see how to scale up the app and run multiple copies of it on this single instance.

In the process, we'll learn about the additional tools Docker provides for creating infrastructure, and deploying and managing production apps. Although we're working locally, the skills will translate directly to doing these same things on cloud-based infrastructure.

Creating Machines

If we're going to deploy our application in production, we need to have somewhere to deploy it to. That means, infrastructure: machine instances capable of running code.

As we saw in Chapter 11, The Production Landscape, on page 129, there are a number of tools available for creating and configuring our infrastructure. However, since this is a book about Docker, we're going to stick with Docker's own tool—Docker Machine—for handling this.